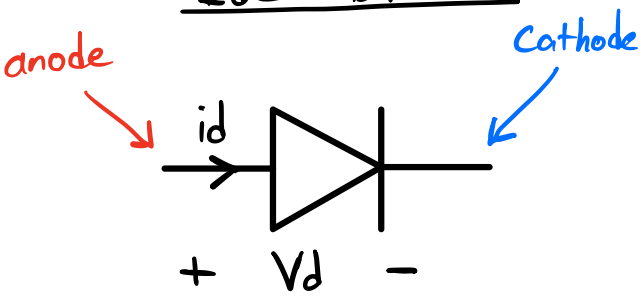
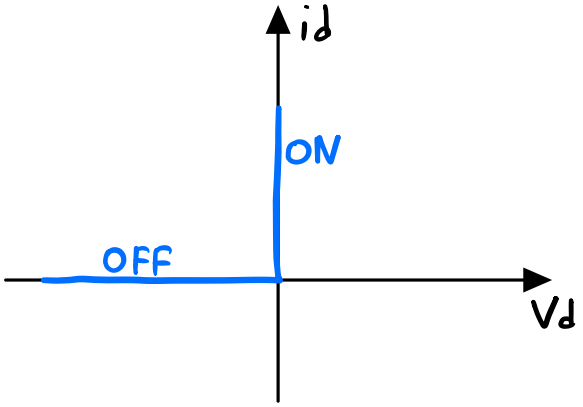
Diodes

Ideal Diode



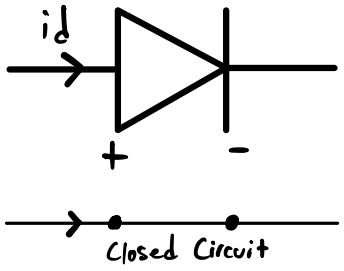
- V_F (V forward) = 0 (in ideal)

Characteristic Curve (Ideal Case):



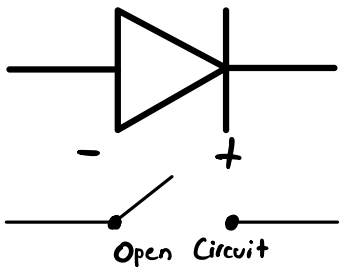
Ideal Diode (Modes of Operation):

Forward Biased

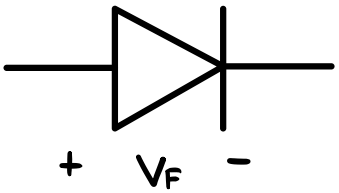


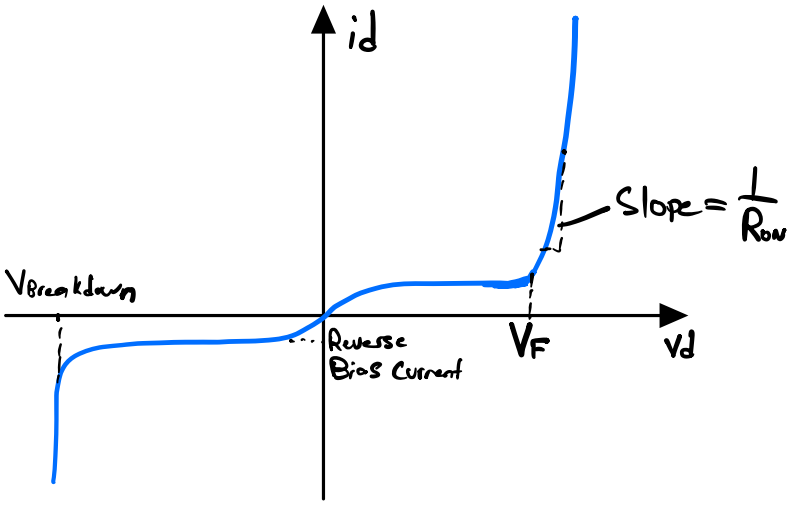
- Ideal Scenario, time it takes to switch between these states $\Delta t = 0$. (Only in Ideal)

Reverse Biased



Practical Diode





- Forward Voltage. Voltage drop across the diode when its conducting and current is flowing in the positive direction

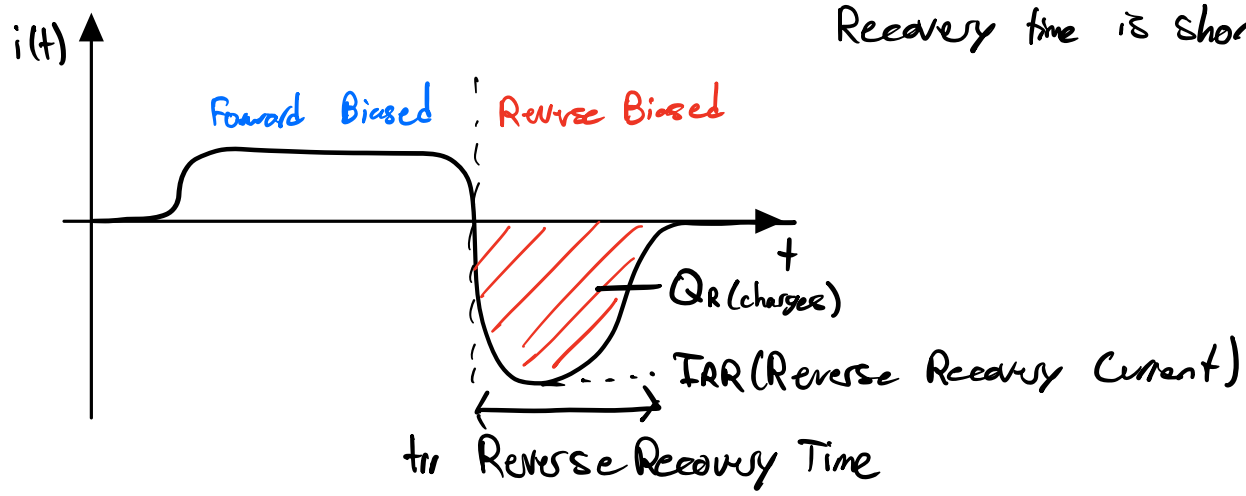
- Reverse-Break-Down Voltage. The voltage when the diode is reverse biased where it will begin conducting in the reverse direction (ie: 1000V) - Damages diode

- On-Resistance

- Turn on/off times

Diode Turn off Time:

- Fast Recovery Diodes, Reverse Recovery time is shorter



Losses

Conduction → $P_{cond} = V_F \cdot I_F \cdot D$ (Dominant at Low Frequencies)

$P_{cond} = I_D^2 \cdot R_{DSON}$

Switching → (Increases with turn ON/turn OFF time)

(Increases with Switching Frequency)

P_{SW} = (Losses per switch (Q))(Time of one period)

ie: How many switches per second